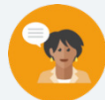


7.7 The Unit Circle and Its Coordinates

Lesson Introduction

	Benchmarks	MA.912.T.1.1 Define trigonometric ratios for acute angles in right triangles.		MA.912.T.1.2 Solve mathematical and real-world problems involving right triangles using trigonometric ratios and the Pythagorean Theorem.			Learning Targets	- I can explore relationships between the coordinates of vertices when the triangle sides are in different quadrants. - I can use side ratios of 45-45-90 triangles to solve problems. - I can use side ratios of 30-60-90 triangles to solve problems.	
	Benchmark Coherence	Building On					Working Toward		
		MA.8.GR.1.1 Apply the Pythagorean Theorem to solve mathematical and real-world problems involving unknown side lengths in right triangles.					N/A		
	Essential Questions	How do I identify coordinates on the unit circle?							
	Lesson Narrative	This lesson is the first experience students have with the unit circle on a polar graph. Students will work in groups to discover the coordinates on the unit circle by making connections to the side relationships of special right triangles.							
	Component Summary	7.7.1 Warm-Up (~5 min.)	Determining the time for two things to cross Florida (SE p. 408)		7.7.3 Bring It Together (5-10 min.)	Summarizing the findings from the Unit Circle (SE p. 414)			
		7.7.2 Exploration (20-25 min.)	Identifying the coordinates on the Unit Circle (SE p. 408)		7.7.4 Try It (5 min.)	Finding coordinates using the Unit Circle (SE p. 415)			
		7.7.5 Wrap-Up (~5 min.)	Reflecting on understanding and confusions over the Unit Circle (SE p. 415)						
	Resources	Glossary		Materials		Additional Preparation			
		<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Unit Circle - Polar Graph		(Optional) Colored pencils (Optional) Highlighters (Optional) Colored tape		If creating a visual Unit Circle before the lesson, tape the large Unit Circle in a central location in the classroom.			

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may incorrectly identify the x- and y-coordinates or use the incorrect sign. - Students may struggle with working with radicals and recognizing equivalent expressions.	Some calculators may rationalize the denominators in the fractions involving radicals. Consider allowing this lesson to be a non-calculator lesson in order to avoid the confusion, and to monitor student understanding of sine, cosine, and tangent in special right triangles.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder Students will engage in identifying similarities and differences in question 2a during the 7.7.2 Exploration. Consider adapting this to have students first engage in the Notice and Wonder routine. During 7.7.3 Bring It Together, prompt students with “What do you notice? What do you wonder?” about the triangles shown. Allow students time to think and to write a response before spotlighting different responses.	MA.K12.MTR.5.1: Patterns and Structure The 7.7.2 Exploration is scaffolded for students to write conjectures as they notice patterns and structure in coordinates on a polar graph. Students will bring together their understanding from the exploratory activity in the 7.7.3 Bring It Together, where students will apply the patterns they observed in the 7.7.2 Exploration to find unknown measures and unknown coordinates.
 Differentiate Instruction	Consider taping a large unit circle in a central location where students can identify different angle measures and label the coordinates on the circle. This will allow students to visually see the information while using movement to identify their solution. Throughout the lesson, use question stems to help auditory learners have an entry point when answering questions.	
Lesson Notes:		

7.7.1 Warm-Up: Between Two Numbers

Component Overview: Using the information given, students will determine how long it will take a turtle and a drone to travel across Florida.



7.7.1 Warm-Up: Between Two Numbers



After observing student work, highlight 2-3 students who have solved the problems differently and discuss how each approach is correct.

1. A Florida Box Turtle's average speed is about 0.17 miles per hour.

The North American X-15's top speed is about 163 miles per hour.

How much time would it take for each to travel across Florida, a distance of about 500 miles?

It would take the box turtle about 2941.176 hours or 122.5 days and 3.067 hours.

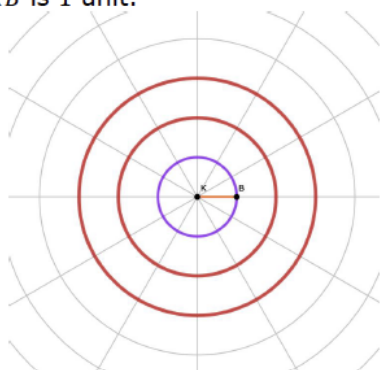
7.7.2 Exploration: The Unit Circle

Component Overview: Students will explore the coordinates of the unit circle using the sine, cosine, and tangent ratios for special right triangles.

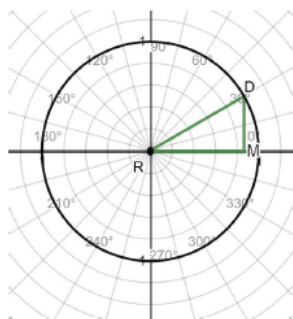
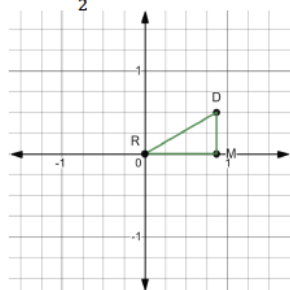


7.7.2 Exploration: The Unit Circle

- Circle K is shown on polar graph paper, where the length of $\overline{KB}$ is 1 unit.



- Draw a circle that has a radius of 2 units.
Answer shown on polar graph.
 - Draw a circle that has a radius of 3 units.
Answer shown on polar graph.
- $\triangle RDM$ is a right triangle formed by drawing a 30° angle at the origin. $\triangle RDM$ is shown on both a coordinate grid and on a polar graph, where $m\angle MRD = 30^\circ$, $m\angle RDM = 60^\circ$, $m\angle DMR = 90^\circ$, $RD = 1$, $DM = \frac{1}{2}$, and $RM = \frac{\sqrt{3}}{2}$.



This Exploration is scaffolded to allow students to work through the problems with their peers. Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share in the whole-group 7.7.3 Bring It Together.



Using no more than 2 minutes, facilitate a discussion regarding the difference between a polar graph and a coordinate graph. Allow students to discuss when a polar graph is useful in real-life scenarios but be mindful of timing.

7.7.2 Exploration: The Unit Circle (continued)

- a. Discuss with your partner any similarities and differences between the graphs. Summarize your discussion.

Answers will vary.

Both graphs have a 30 – 60 – 90 triangle. One is a coordinate grid and the other is a polar graph.

- b. Work with your partner to determine the coordinates of point D .

$$\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$$

- c. Work with your partner to complete the table.
Leave answers in exact form.

Angle	Cosine	Sine	Tangent
30° ($m\angle R$)	$\frac{\sqrt{3}}{2}$ $\frac{1}{1}$	$\frac{1}{2}$ $\frac{2}{1}$	$\frac{1}{2}$ $\frac{\sqrt{3}}{2}$

- d. Discuss with your partner how the coordinates of point D can be found using the values from the table. Make a conjecture about the relationship between the values of cosine and sine, and the coordinate pair formed by the angle.

Answers will vary. Using the angle located at the origin as the reference angle, the cosine of the reference angle represents the x value of the coordinate point and the sine represents y value of the coordinate point. (cosine, sine)



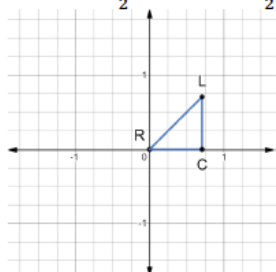
Consider posting a large unit circle with different angle measures listed on the circle. Students can use different tape colors or markings to create the triangles around the center and label the coordinates along the outside of the circle.



How do the side lengths of $\triangle RDM$ relate to special right triangles?

7.7.2 Exploration: The Unit Circle (continued)

3. $\triangle RLC$ is a right triangle formed by drawing a 45° angle at the origin. $\triangle RLC$ is shown on both a coordinate grid and on a polar graph, where $m\angle CRL = 45^\circ$, $m\angle RLC = 45^\circ$, $m\angle RCL = 90^\circ$, $RL = 1$, $RC = \frac{\sqrt{2}}{2}$, and $LC = \frac{\sqrt{2}}{2}$.



- a. Determine the coordinates of point L .

$$\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$$

- b. Work with your partner to complete the table.
Leave answers in exact form.

Angle	Sine	Cosine	Tangent
45° ($m\angle R$)	$\frac{\sqrt{2}}{2}$ $\frac{2}{1}$	$\frac{\sqrt{2}}{2}$ $\frac{2}{1}$	$\frac{\sqrt{2}}{2}$ $\frac{2}{2}$

- c. Discuss with your partner whether or not the conjecture that was made about the coordinates of point D could be applied to determine the coordinates of point L . If necessary, revise your conjecture.

Answers may vary. Yes, it can be used. The coordinate point of D and L are both using (cosine, sine).

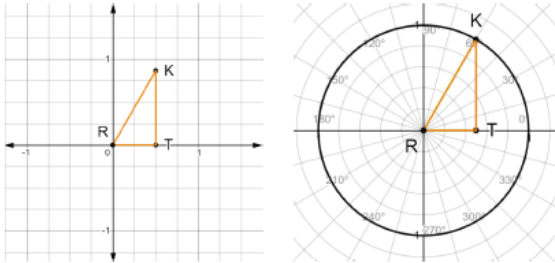


Discussion provides additional entry points for auditory learners. Consider utilizing sentence stems and provide ample time for students to grapple with academic language:

- _____ represents the ____-coordinate, while _____ represents the ____-coordinate.

7.7.2 Exploration: The Unit Circle (continued)

4. $\triangle RKT$ is a right triangle formed by drawing a 60° angle at the origin. $\triangle RKT$ is shown on both a coordinate grid and on a polar graph, where $m\angle TRK = 60^\circ$, $m\angle RKT = 30^\circ$, $m\angle RTK = 90^\circ$, $RK = 1$, $RT = \frac{1}{2}$, and $TK = \frac{\sqrt{3}}{2}$.



- a. Determine the coordinates of point K .

$$\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$$

- b. Work with your partner to complete the table.
Leave answers in exact form.

Angle	Sine	Cosine	Tangent
60° ($m\angle R$)	$\frac{\sqrt{3}}{2}$ $\frac{2}{1}$	$\frac{1}{2}$ $\frac{2}{1}$	$\frac{\sqrt{3}}{2}$ $\frac{1}{2}$

- c. Discuss with your partner how the coordinates will change if the triangle is in a different quadrant. Make a prediction.
Answers will vary. The coordinates will become positive or negative depending on the quadrant.



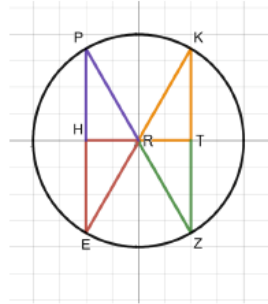
Challenge students to justify why the cosine and sine could be written as $\frac{1}{\sqrt{2}}$ and as $\frac{\sqrt{2}}{2}$, and how this relates to the side relationships of 45-45-90 triangles.



What are the similarities and differences between the triangle in question 4 and the triangle in question 2?

7.7.2 Exploration: The Unit Circle (continued)

5. Four triangles were formed by drawing angles of 60° from the origin. $m\angle TRK = 60^\circ$, $m\angle PRH = 60^\circ$, $m\angle HRE = 60^\circ$, and $m\angle TRZ = 60^\circ$. $RK = 1$, $RP = 1$, $RE = 1$, and $RZ = 1$.



- a. Complete the table.

Point	Coordinates
K	$\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$
P	$\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$
E	$\left(-\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$
Z	$\left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$

- b. Points K , P , E , and Z lie on circle R . What is the radius of circle R ?

1

- c. What type of special triangles are ΔRKT , ΔRPH , ΔREH , and ΔRZT ?

30 – 60 – 90



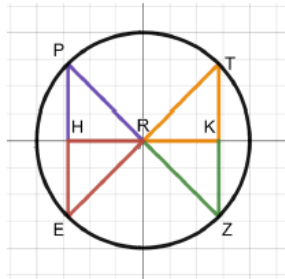
How are sine and cosine related in ΔRDM and ΔKTR ? Why do you think that is?



For kinesthetic and visual learners, give students two different 30-60-90 triangles, and one 45-45-90 triangle. Students can write the side lengths on the triangles with a hypotenuse of 1 and move it around the circle while finding the coordinates.

7.7.2 Exploration: The Unit Circle (continued)

6. Four triangles were formed by drawing angles of 45° from the origin. $m\angle TRK = 45^\circ$, $m\angle PRH = 45^\circ$, $m\angle HRE = 45^\circ$, and $m\angle KRZ = 45^\circ$. $RT = 1$, $RP = 1$, $RE = 1$, and $RZ = 1$.



- a. Complete the table.

Point	Coordinates
T	$\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$
P	$\left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$
E	$\left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$
Z	$\left(\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$

- b. Points T , P , E , and Z lie on circle R . What is the radius of circle R ?

1

- c. What type of special triangles are $\triangle RTK$, $\triangle RPH$, $\triangle REH$, and $\triangle RKZ$?

45 – 45 – 90

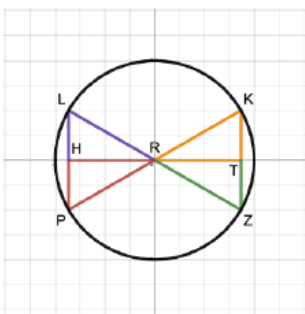


Common Misconception

Students may swap the x - and y -coordinate when listing the coordinates of the points.

7.7.2 Exploration: The Unit Circle (continued)

7. Four triangles were formed by drawing angles of 30° from the origin. $m\angle TRK = 30^\circ$, $m\angle PRH = 30^\circ$, $m\angle HRL = 30^\circ$, and $m\angle TRZ = 30^\circ$. $RK = 1$, $RP = 1$, $RL = 1$, and $RZ = 1$.



Students may find the coordinates to be variations of $(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$, which are equivalent coordinates.

- a. Complete the table.

Point	Coordinates
K	$(\frac{\sqrt{3}}{2}, \frac{1}{2})$
L	$(-\frac{\sqrt{3}}{2}, \frac{1}{2})$
P	$(-\frac{\sqrt{3}}{2}, -\frac{1}{2})$
Z	$(\frac{\sqrt{3}}{2}, -\frac{1}{2})$

- b. Points K , L , P , and Z lie on circle R . What is the radius of circle R ?

1

- c. What type of special triangles are ΔRKT , ΔRPH , ΔRLH , and ΔRZT ?

30 – 60 – 90

7.7.2 Exploration: The Unit Circle (continued)

8. For each type of special triangle, what similarities do the coordinates for each vertex located on the circle share with one another?

Answers will vary. The values of the coordinates all have the same absolute value.

9. For each type of special triangle, what differences do the coordinates for each vertex located on the circle share with one another?

Answers will vary. The coordinates have different signs depending on their location on the coordinate grid.



Notice and Wonder

Before students answer questions 8 and 9, consider utilizing a “Notice and Wonder” routine to offers students the opportunity to make observations regarding the coordinates and to talk through the task to solidify their thinking.

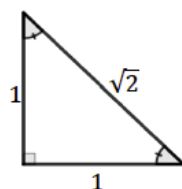
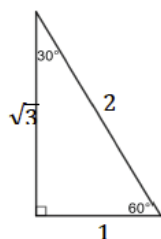
7.7.3 Bring It Together: The Unit Circle

Component Overview: Using the findings from 7.7.2 Exploration, the teacher will discuss how to use the circle to find the coordinates on the circle.



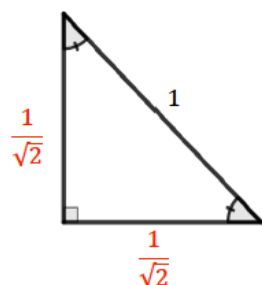
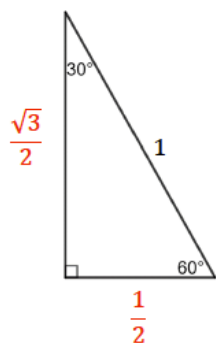
7.7.3 Bring It Together: The Unit Circle

In the Exploration, all of the triangles were one of the two special triangles, $30^\circ - 60^\circ - 90^\circ$ or $45^\circ - 45^\circ - 90^\circ$. The relationships of each of the side lengths for these triangles are shown.



In the circle, the hypotenuse of the triangle was the radius of the circle, which was 1. When a circle has a radius of 1 and is centered at the origin, it is called the unit circle. In the diagrams of the two special triangles the hypotenuse is not 1.

1. Fill in the blanks to show the relationships of the special triangles when the hypotenuse is 1.

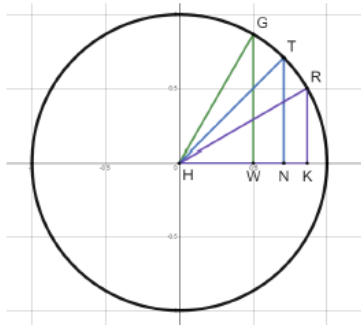


Notice and Wonder

Prompt students with, “What do you notice? What do you wonder?” about the triangles shown. Allow students time to think and to write a response before spotlighting different responses.

7.7.3 Bring It Together: The Unit Circle (continued)

2. A unit circle with $\triangle HKR$, $\triangle HGW$, and $\triangle HNT$ is shown, where $m\angle KHR = 30^\circ$, $m\angle NHT = 45^\circ$, $m\angle WHG = 60^\circ$, $m\angle GWH = 90^\circ$, $m\angle HNT = 90^\circ$, and $m\angle HKR = 90^\circ$.



Allow students to use colored pencils and/or highlighters to color code each reference angle.

Complete the table.

Point	Coordinates
G	$\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$
R	$\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$
T	$\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$

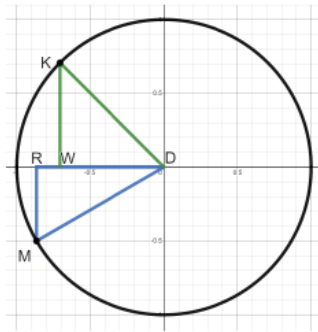
7.7.4 Try It: Finding Coordinates in the Unit Circle

Component Overview: Students work independently to practice using the Unit Circle to find coordinates based on a given reference angle of 30° , 45° , or 60° .



7.7.4 Try It: Finding Coordinates in the Unit Circle

1. A unit circle is shown where $\triangle KDW$ and $\triangle MRD$ are right triangles, $m\angle KDW = 45^\circ$, and $m\angle RDM = 30^\circ$.



Complete the table.

Point	Coordinates
K	$\left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$
M	$\left(-\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$



Common Misunderstandings

If the signs are incorrect for the coordinates, consider facilitating a discussion over the x - and y -coordinates based on the quadrants. Students can review graphing using On-Ramp to Algebra 1, “Graphing Ordered Pairs.”

If the coordinates of point M are swapped, then the students have a misunderstanding between a 30° and 60° reference angle.

7.7.5 Wrap-Up: Reflection

Component Overview: Students will reflect on their understanding and wonderings from the exploration at the start of the lesson.



7.7.5 Wrap-Up: Reflection

1. Complete the table.

Answers will vary.

The terms and information used in the Exploration that I knew were ...	Something I wondered during the Exploration was ...	Something I learned during the Exploration was ...



Consider what students write in each column to determine what remediation steps may be needed.